NATIONAL INSTITUTE OF TECHNOLOGY, RAIPUR
Department of Information Technology — Minor Project Defense & Visual Guide

SGSCF-Net Presentation Defense Script with Slide Visuals

Spatial-Guided Spectral-Channel Fusion Network for Hyperspectral Remote Sensing Image Analysis

Supervisor: Dr. R.R. Janghel, Associate Professor | **Presenters:** Tejas Surya (23118098), B. Vyshnavi (23118021), Valluri Lahitha (23118106)

EXECUTIVE PRESENTATION STRATEGY & NUMERICAL TRUTH SHEET

Presentation Structure & Team Division (15 Minutes Total):

• **Speaker 1 (Tejas Surya) — Slides 1 to 7 (5.0 min):** Title, Outline, Remote Sensing Fundamentals, HSI Principles, Imaging Comparison, Applications.

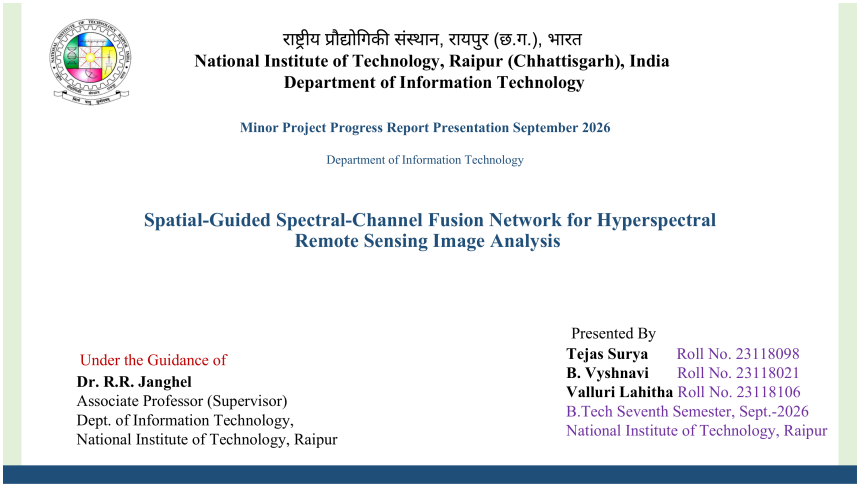
• **Speaker 2 (B. Vyshnavi) — Slides 8 to 13 (5.0 min):** Benchmark Datasets, Feature Extraction Workflow, Deep Learning Limitations, Literature Review (Parts 1 & 2), Research Gaps.

• **Speaker 3 (Valluri Lahitha) — Slides 14 to 21 (5.0 min):** Empirical Flaws & Bottlenecks, Research Objectives, Techniques Employed, Proposed Architecture Workflow, Phase-wise Research Plan, Experimental Sweeps, References, Conclusion.

Metric / Benchmark	Indian Pines (AVIRIS)	Pavia University (ROSIS)	Salinas Valley (AVIRIS)
Spatial Grid / GSD	145 × 145 px, 20m/pixel	610 × 340 px, 1.3m/pixel	512 × 217 px, 3.7m/pixel
Spectral Channels	200 calibrated bands	103 spectral channels	204 calibrated bands
SGSCF-Net OA (%)	**98.05%**	**99.82%**	**99.92%**
SGSCF-Net AA (%)	**94.82% (RANK 1 SOTA)**	**99.71%**	**99.92%**
Minority Accuracy	Oats: 54.17% (SSRN: 0.0%)	Gravel: 99.76% (SSAN: 17.9%)	11 of 16 classes at 100.0%
Parameters & Checkpoint	**308,097 params (1.2 MB)**	3 Attention modules = 3.37%	vs SSAN: 6.40M (20.8× smaller)

SLIDE 1 / 21: TITLE SLIDE & PROJECT OVERVIEW

Time: 45 Seconds | **Speaker:** Tejas Surya | **Visual Focus:** *NIT Raipur bilingual banner, Project title, Dr. R.R. Janghel, Student roll numbers.*



WHAT YOU MUST SAY (Point-Wise Spoken Script)

- **Opening Greeting:** 'Respected supervisor Dr. R.R. Janghel sir, and distinguished members of the evaluation committee, a very good morning.'
- **Project Declaration:** 'Today, on behalf of our team comprising Tejas Surya, B. Vyshnavi, and Valluri Lahitha, I present our Minor Project: *Spatial-Guided Spectral-Channel Fusion Network for Hyperspectral Remote Sensing Image Analysis* (SGSCF-Net).'
- **Core Motivation:** 'Hyperspectral imaging captures hundreds of narrow, continuous spectral bands, enabling material identification with spectroscopic accuracy. However, existing deep models suffer from severe overfitting, heavy compute exceeding 85 million parameters, and minority class collapse.'
- **Project Goal:** 'In this presentation, we outline the foundational theory, systematic literature analysis, architectural bottlenecks, and our proposed SGSCF-Net design of only 308K parameters that resolves these challenges.'

KEY CONCEPTS & PRONUNCIATION GUIDE

- **SGSCF-Net Pronunciation:** 'S-G-S-C-F Net' (Spatial-Guided Spectral-Channel Fusion Network).
- **Hyperspectral (HSI):** Imaging across contiguous narrow wavelengths ($400 - 2500$ nm).

VIVA DEFENSE SHIELD (Anticipated Examiner Questions & Winning Answers)

Examiner: What is the primary motivation for selecting Hyperspectral Image Classification?

Winning Defense Answer: HSI bridges 2D visual imagery with laboratory spectroscopy. Unlike RGB imagery which only captures 3 broad bands, HSI captures 100 to 200+ narrow continuous bands, enabling identification of crop species, mineral compositions, and urban materials that appear identical to the naked eye. However, high dimensionality combined with limited labeled data makes it an important research challenge.

SLIDE 2 / 21: PRESENTATION OUTLINE

Time: 30 Seconds | **Speaker:** Tejas Surya | **Visual Focus:** *9-point structured presentation agenda.*

Outline

» Introduction
» Literature Review
» Datasets & Workflows
» Emprical Flaws Identified
» Research Gaps Identified
» Techniques Employed
» Proposed Architecture & Research Plan
» Experimental Plan & Expected Outcomes
» References

WHAT YOU MUST SAY (Point-Wise Spoken Script)

- **Agenda Walkthrough:** 'Here is the roadmap for our presentation today.'
- **Foundations to Review:** 'We begin with an introduction to remote sensing and hyperspectral principles, followed by a comprehensive review of 8 state-of-the-art literature methods.'
- **Bottlenecks to Architecture:** 'Next, we examine the empirical flaws and research gaps identified, followed by our proposed techniques: SBAM, CBSA, SCFG, and our Spread-Focal loss.'
- **Research Execution:** 'Finally, we detail our proposed architecture workflow, phase-wise research milestones, experimental sensitivity plan, and references.'

KEY CONCEPTS & PRONUNCIATION GUIDE

- **Terminology:** SOTA = State-of-the-Art; Empirical Bottlenecks = Concrete failure modes observed in existing benchmarked codebases.

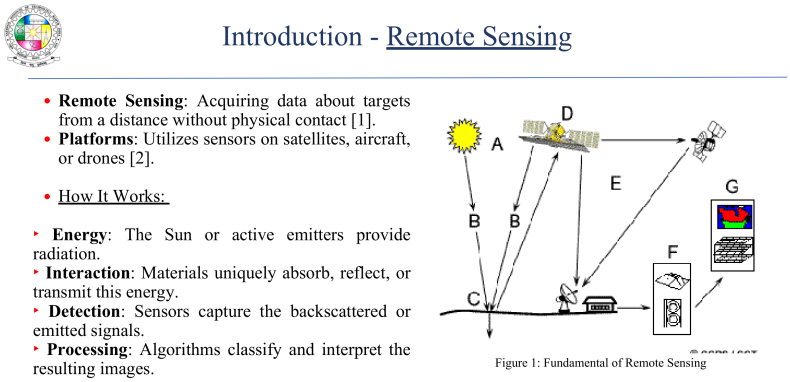
VIVA DEFENSE SHIELD (Anticipated Examiner Questions & Winning Answers)

Examiner: Why is a thorough literature review essential before proposing an architecture?

Winning Defense Answer: To avoid incremental guesswork. By systematically categorizing 3D-CNNs, Transformers, State Space Models, and Capsule networks, we isolate the exact mathematical causes of failure—such as hardware locks, static tensor crashes, and minority collapse—and design specific mathematical modules to solve each.

SLIDE 3 / 21: INTRODUCTION - REMOTE SENSING

Time: 45 Seconds | **Speaker:** Tejas Surya | **Visual Focus:** *Figure 1: Energy radiation, material interaction, sensor capture, algorithm processing.*



WHAT YOU MUST SAY (Point-Wise Spoken Script)

- **Definition:** 'Remote sensing is the science of acquiring physical and chemical information about the Earth's surface from a distance, without direct physical contact.'
- **Sensor Platforms:** 'Data is collected across multiple platforms: low-Earth orbit satellites, airborne aircraft, and uncrewed aerial vehicles (UAVs or drones).'
- **Four-Step Physical Process:** 'The process follows four stages: (1) An energy source—typically the Sun—emits electromagnetic radiation; (2) The radiation interacts with Earth surface materials, undergoing wavelength-specific absorption and reflection; (3) Optical sensors capture the backscattered radiance; (4) Deep learning algorithms classify the pixels into land-cover maps.'
- **Significance:** 'Remote sensing enables macro-scale monitoring of agriculture, forestry, disaster management, and urban expansion.'

KEY CONCEPTS & PRONUNCIATION GUIDE

- **Passive vs. Active Sensing:** Passive sensors measure reflected solar radiance (L_λ); active sensors (like LiDAR/Radar) emit their own coherent pulses.

VIVA DEFENSE SHIELD (Anticipated Examiner Questions & Winning Answers)

Examiner: What type of remote sensing is hyperspectral imaging?

Winning Defense Answer: HSI is primarily passive optical remote sensing, operating in the Solar Reflective region from the visible (400 nm) through Near-Infrared (NIR, 700-1000 nm) to Short-Wave Infrared (SWIR, 1000-2500 nm).

SLIDE 4 / 21: HYPERSPECTRAL IMAGING (HSI)

Time: 50 Seconds | **Speaker:** Tejas Surya | **Visual Focus:** *Figure 2: Hypercube (X, Y, Lambda) and continuous spectral reflection curves.*

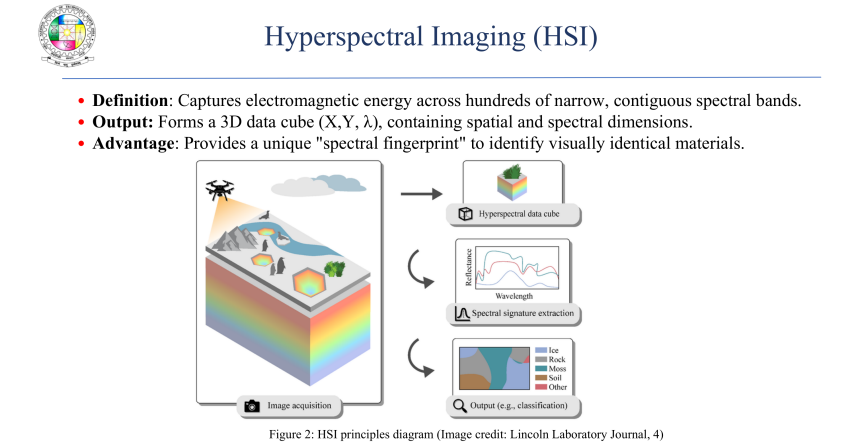


Figure 2: HSI principles diagram (Image credit: Lincoln Laboratory Journal, 4)

WHAT YOU MUST SAY (Point-Wise Spoken Script)

- **Core Definition:** 'Hyperspectral Imaging (HSI) captures electromagnetic energy across hundreds of narrow, contiguous spectral bands, typically between 100 and 300+ channels.'
- **The 3D Data Cube:** 'The sensor output is a 3D volume or data cube of dimensions (X, Y, λ), where X and Y are spatial coordinates, and λ represents the continuous spectral wavelength dimension.'
- **The Spectral Fingerprint:** 'Every spatial pixel in this cube contains a continuous spectral reflectance curve. Because different materials have distinct chemical bonds, their absorption features act as unique spectral fingerprints.'
- **Key Advantage:** 'This allows us to separate materials that look visually identical in conventional RGB photography—such as distinguishing different tree species, plastics, or crop maturity stages.'

KEY CONCEPTS & PRONUNCIATION GUIDE

- **Data Cube Coordinates:** $X(x, y, \lambda)$ where (x,y) are spatial 2D coordinates and $\lambda \in [\lambda_{min}, \lambda_{max}]$ is the wavelength coordinate.

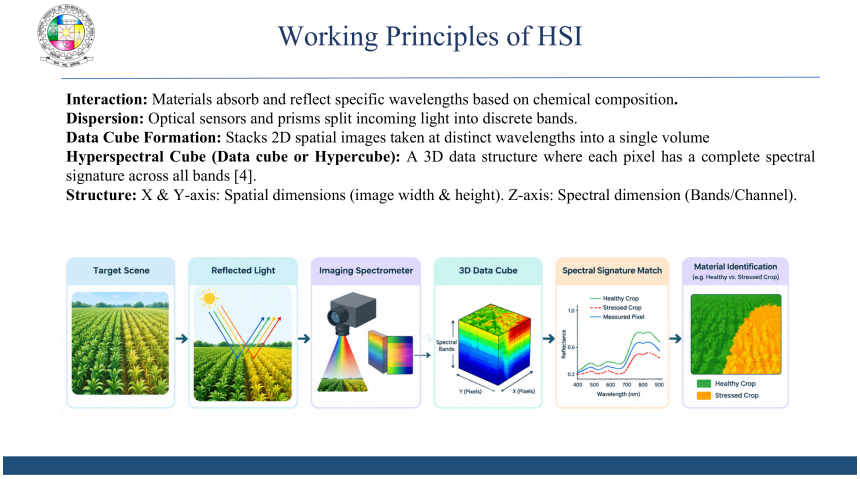
VIVA DEFENSE SHIELD (Anticipated Examiner Questions & Winning Answers)

Examiner: What is a spectral signature and what causes it?

Winning Defense Answer: A spectral signature is the reflectance percentage of an object as a function of wavelength. It is governed by molecular vibrational overtones and electronic transitions in chemical bonds (e.g., chlorophyll absorption in the red band, water absorption in the SWIR).

SLIDE 5 / 21: WORKING PRINCIPLES OF HSI

Time: 50 Seconds | **Speaker:** Tejas Surya | **Visual Focus:** *Prism optical dispersion diagram, band splitting, and 3D volume stacking.*



WHAT YOU MUST SAY (Point-Wise Spoken Script)

- **Physical Dispersion:** 'How does an HSI sensor physically generate this data cube? Incident light reflected from a target enters an imaging spectrometer.'
- **Optical Splitting:** 'An internal dispersing element—such as a diffraction grating or prism—splits the incoming broadband light into discrete, narrow wavelengths.'
- **2D Detector Array:** 'A linear slit projects one spatial line across the detector while the orthogonal axis captures the dispersed spectral bands, scanning row-by-row in a pushbroom manner.'
- **Hypercube Assembly:** 'These stacked 2D spatial frames at contiguous wavelengths form the 3D hypercube. X and Y provide spatial context, while the Z-axis provides rich spectroscopic depth.'

KEY CONCEPTS & PRONUNCIATION GUIDE

- **Pushbroom Scanning:** Captures an entire spatial line simultaneously, dispersing wavelengths across a 2D focal plane array (FPA).

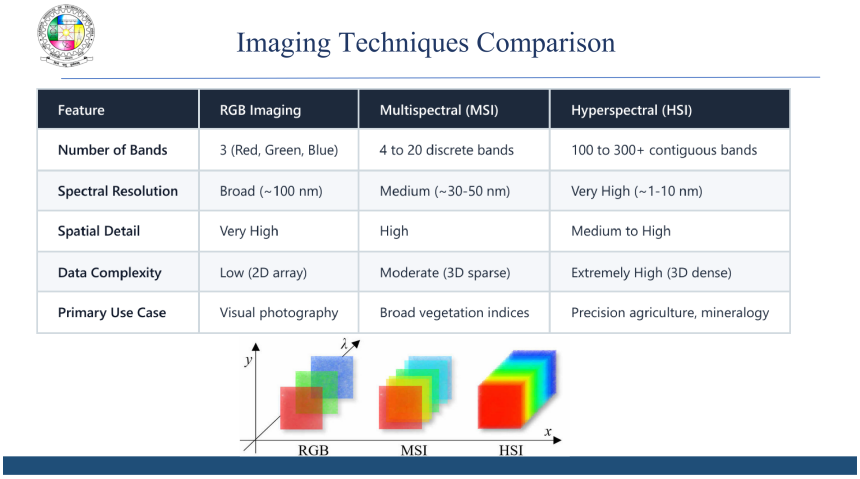
VIVA DEFENSE SHIELD (Anticipated Examiner Questions & Winning Answers)

Examiner: Why does the 3D structure of an HSI cube create challenges for standard 2D convolutions?

Winning Defense Answer: Standard 2D convolutions collapse all spectral channels into a single depth sum, treating them like arbitrary color channels. This obliterates continuous spectral curvature and inter-band absorption slopes. True spectral preservation requires 3D convolutions or targeted spectral attention.

SLIDE 6 / 21: IMAGING TECHNIQUES COMPARISON

Time: 60 Seconds | **Speaker:** Tejas Surya | **Visual Focus:** *Comparative table: RGB vs Multispectral (MSI) vs Hyperspectral (HSI) vs Ultraspectral.*



WHAT YOU MUST SAY (Point-Wise Spoken Script)

- **Comparative Spectrum:** 'This slide provides a clear comparison between RGB imaging, Multispectral Imaging (MSI), and Hyperspectral Imaging (HSI).'
- **Number of Bands:** 'RGB has only 3 broad bands (Red, Green, Blue). MSI captures 4 to 20 discrete bands. HSI captures 100 to 300+ contiguous bands.'
- **Spectral Resolution:** 'RGB has broad resolution (~100 nm). MSI has medium resolution (~30–50 nm) with wide unmeasured gaps. HSI has high resolution (~1–10 nm) contiguous sampling.'
- **Data Complexity:** 'RGB is a simple 2D array; MSI is moderate; HSI is an extremely dense 3D volume requiring high-performance processing.'
- **Use Case Distinction:** 'RGB is limited to visual inspection. MSI is suitable for broad vegetation indices like NDVI. HSI is required for precision agriculture, mineralogy, and chemical differentiation.'

KEY CONCEPTS & PRONUNCIATION GUIDE

- **Spectral Sampling Interval:** $\Delta \lambda \le 10\text{ nm}$ in HSI ensures continuous absorption line capture.

VIVA DEFENSE SHIELD (Anticipated Examiner Questions & Winning Answers)

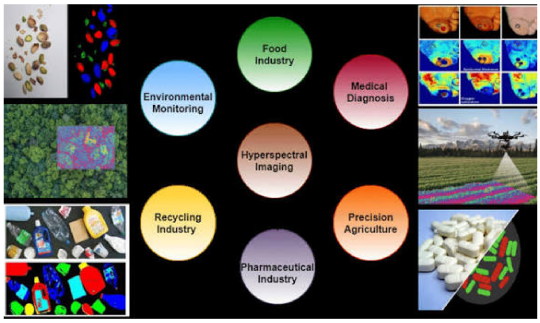
Examiner: Why can't Multispectral Imaging (like Sentinel-2) identify subtle crop diseases?

Winning Defense Answer: Multispectral sensors have wide spectral gaps between bands (e.g., 50 to 100 nm). Early crop stress causes subtle shifts in the 'red edge' slope (690–740 nm) and water absorption depth at 970 nm, which fall completely between MSI bands. HSI samples continuously at 10 nm intervals, detecting stress weeks earlier.

SLIDE 7 / 21: APPLICATIONS OF HYPERSPECTRAL IMAGING

Time: 45 Seconds | **Speaker:** Tejas Surya | **Visual Focus:** *Application chart: Agriculture, Medical, Environment, Recycling, Food, Pharmaceuticals.*

Applications of HSI:



WHAT YOU MUST SAY (Point-Wise Spoken Script)

- **Multidisciplinary Scope:** 'Hyperspectral imaging has revolutionized diverse scientific and industrial fields, shown here in six key sectors.'
- **Precision Agriculture:** 'In agriculture, HSI detects nitrogen deficiency, soil salinity, moisture stress, and weed infestations at the single-plant level.'
- **Environmental & Mineralogy:** 'It enables automated mapping of mineral deposits, tracking oil spill dispersion, and quantifying greenhouse gas plumes.'
- **Medicine & Industry:** 'In medicine, HSI assists in non-invasive cancer tissue margin detection; in manufacturing, it inspects pharmaceutical tablet uniformity and sorts recyclable plastics by polymer type.'
- **Our Project Focus:** 'Our work focuses on precision agriculture and urban scene classification, validated across standard international benchmarks.'

KEY CONCEPTS & PRONUNCIATION GUIDE

- **Biochemical Indicators:** Chlorophyll absorption (670 nm), Carotenoids (450 nm), Water absorption ($970, 1200, 1450\text{ nm}$).

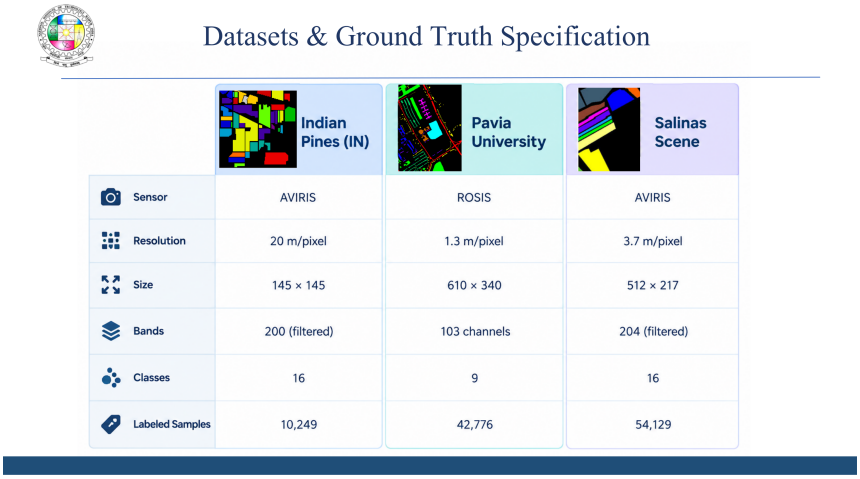
VIVA DEFENSE SHIELD (Anticipated Examiner Questions & Winning Answers)

Examiner: Which specific datasets in your study correspond to these applications?

Winning Defense Answer: Indian Pines and Salinas Valley represent precision agriculture (crop type and maturity stage classification). Pavia University represents urban infrastructure and environmental mapping (differentiating asphalt, gravel, bitumen, and vegetation).

SLIDE 8 / 21: DATASETS & GROUND TRUTH SPECIFICATION

Time: 60 Seconds | **Speaker:** B. Vyshnavi | **Visual Focus:** *Dataset table: Indian Pines, Pavia University, Salinas Scene parameters.*



WHAT YOU MUST SAY (Point-Wise Spoken Script)

- **Dataset Overview:** 'Thank you, Tejas. I will now present our benchmark datasets and experimental workflows.'
- **Indian Pines (AVIRIS):** 'Captured by AVIRIS over Northwest Indiana. Spatial resolution is 20m/pixel across a 145×145 grid. It has 200 calibrated spectral bands (20 atmospheric bands removed), 16 agricultural classes, and 10,249 labeled samples. It features extreme class imbalance, from 954 samples down to only 20 samples for Oats.'
- **Pavia University (ROSIS):** 'Captured by ROSIS over Pavia, Italy. Spatial resolution is high at 1.3m/pixel across a 610×340 grid, with 103 spectral channels, 9 urban classes, and 42,776 labeled samples.'
- **Salinas Valley (AVIRIS):** 'Captured by AVIRIS over Salinas Valley, California. Spatial resolution is 3.7m/pixel across a 512×217 grid, with 204 calibrated bands, 16 agricultural classes, and 54,129 labeled samples, including fine-grained crop maturity stages.'

KEY CONCEPTS & PRONUNCIATION GUIDE

- **Atmospheric Water Absorption Bands:** AVIRIS bands 104–108 ($1.4\,\mu\text{m}$), 150–163 ($1.9\,\mu\text{m}$), and band 220 are omitted due to near-zero atmospheric transmission.

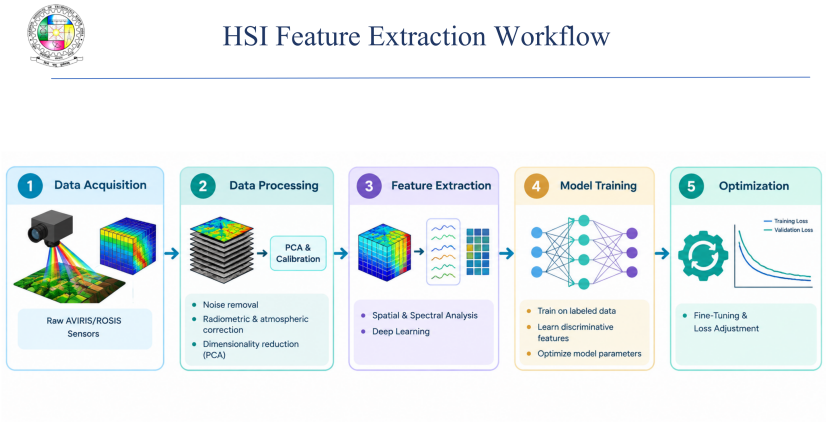
VIVA DEFENSE SHIELD (Anticipated Examiner Questions & Winning Answers)

Examiner: Why is Indian Pines considered the most difficult HSI benchmark?

Winning Defense Answer: Indian Pines combines low spatial resolution (20m/pixel, resulting in severe mixed-pixel effects) with extreme class imbalance: the largest class has 954 samples, while the Oats class has only 20 samples. Most models collapse to 0% on Oats.

SLIDE 9 / 21: HSI FEATURE EXTRACTION WORKFLOW

Time: 45 Seconds | **Speaker:** B. Vyshnavi | **Visual Focus:** *5-step chevron diagram: Acquisition -> Preprocessing -> Feature Extraction -> Training -> Optimization.*



WHAT YOU MUST SAY (Point-Wise Spoken Script)

- **Standard Pipeline:** 'This diagram illustrates the classical HSI processing workflow from raw sensor acquisition to final thematic classification.'
- **Stages 1 & 2 — Acquisition & Preprocessing:** 'Raw sensor radiances are converted to surface reflectance via atmospheric correction. Noisy water absorption bands are removed.'
- **Stages 3 & 4 — Feature Extraction & Training:** 'Spatial and spectral neighborhood patches are extracted and passed to deep learning models to learn discriminative material representations.'
- **Stage 5 — Optimization & Loss:** 'Model weights are iteratively optimized using loss gradients to minimize classification error across all classes.'
- **Limitations of Standard Flow:** 'Traditional pipelines often use linear PCA dimensionality reduction, which discards valuable non-linear spectral signatures. In SGSCF-Net, we replace manual PCA with end-to-end attention.'

KEY CONCEPTS & PRONUNCIATION GUIDE

- **PCA Formulation:** Projects $X \in \mathbb{R}^{N \times C}$ onto principal eigenvectors of covariance $\Sigma = \frac{1}{N}X^TX$, retaining top k variance components.

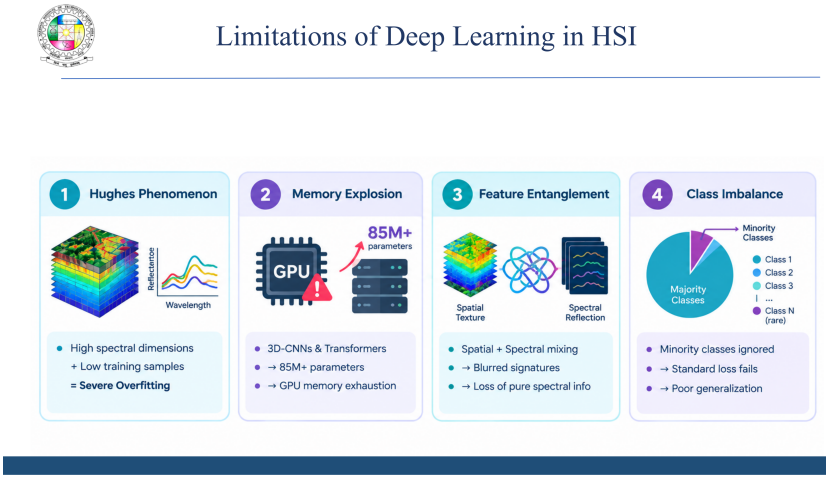
VIVA DEFENSE SHIELD (Anticipated Examiner Questions & Winning Answers)

Examiner: Why does PCA dimensionality reduction sometimes degrade classification?

Winning Defense Answer: PCA is an unsupervised linear projection that maximizes overall variance. However, high-variance bands often correspond to broad illumination changes rather than material differences. Subtle absorption features of rare minority classes have low overall variance and are discarded during PCA truncation.

SLIDE 10 / 21: LIMITATIONS OF DEEP LEARNING IN HSI

Time: 60 Seconds | **Speaker:** B. Vyshnavi | **Visual Focus:** *4 limitation cards: Hughes Phenomenon, Memory Explosion, Feature Entanglement, Class Imbalance.*



WHAT YOU MUST SAY (Point-Wise Spoken Script)

- **Four Major Bottlenecks:** 'Applying standard deep learning directly to hyperspectral data encounters four fundamental bottlenecks.'
- **1. The Hughes Phenomenon:** 'High spectral dimensionality combined with very few labeled training pixels causes exponential parameter explosion and severe overfitting.'
- **2. Memory Explosion:** 'Standard 3D-CNNs and Vision Transformers require tens of millions of parameters (exceeding 85M+ in models like ViT), exhausting GPU VRAM.'
- **3. Feature Entanglement:** 'Simultaneous 3D convolutions mix spatial and spectral features indiscriminately, blurring fine absorption signatures with spatial textures.'
- **4. Severe Class Imbalance:** 'Standard Cross-Entropy loss is dominated by frequent classes, leading to the total collapse of minority classes (0% accuracy on rare crops).'

KEY CONCEPTS & PRONUNCIATION GUIDE

- **Hughes Phenomenon Curve:** Accuracy peaks at optimal dimension d^*, then degrades: $\lim_{d \to \infty} \text{Accuracy} = \text{Random Guess}$ for fixed N samples.

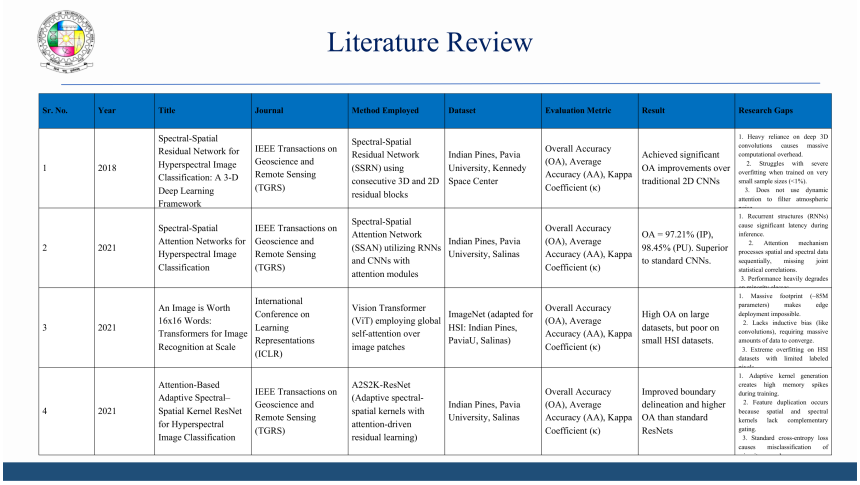
VIVA DEFENSE SHIELD (Anticipated Examiner Questions & Winning Answers)

Examiner: What causes feature entanglement in standard 3D-CNNs?

Winning Defense Answer: Standard 3D convolutions apply kernels of size $K_h \times K_w \times K_d$ that compute weighted sums across both spatial coordinates and spectral channels simultaneously in every layer. This prevents the network from isolating pure spectral chemical signatures independently of local spatial textures.

SLIDE 11 / 21: LITERATURE REVIEW (PART 1: PAPERS 1 TO 4)

Time: 75 Seconds | **Speaker:** B. Vyshnavi | **Visual Focus:** *Table: SSRN (2018), SSAN (2021), ViT (2021), A2S2K-ResNet (2021) with results & research gaps.*



Literature Review

Sr. No.	Year	Title	Journal	Method Employed	Dataset	Evaluation Metric	Result	Research Gaps
1	2018	Spectral-Spatial Residual Network for Hyperspectral Image Classification: A 3-D Deep Learning Framework	IEEE Transactions on Geoscience and Remote Sensing (TGRS)	Spectral-Spatial Residual Network (SSRN) using consecutive 3D and 2D residual blocks	Indian Pines, Pavia University, Kennedy Space Center	Overall Accuracy (OA), Average Accuracy (AA), Kappa Coefficient (κ)	Achieved significant OA improvements over traditional 2D CNNs	1. Heavy reliance on deep 3D convolutions causes massive computational overhead. 2. Struggles with severe overfitting when trained on very small sample sizes (<1%). 3. Does not use dynamic attention to filter atmospheric
2	2021	Spectral-Spatial Attention Networks for Hyperspectral Image Classification	IEEE Transactions on Geoscience and Remote Sensing (TGRS)	Spectral-Spatial Attention Network (SSAN) utilizing RNNs and CNNs with attention modules	Indian Pines, Pavia University, Salinas	Overall Accuracy (OA), Average Accuracy (AA), Kappa Coefficient (κ)	OA = 97.21% (IP), 98.45% (PU). Superior to standard CNNs.	1. Recurrent structures (RNNs) cause significant latency during inference. 2. Attention mechanism processes spatial and spectral data sequentially, missing joint statistical correlations. 3. Performance heavily degrades
3	2021	An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale	International Conference on Learning Representations (ICLR)	Vision Transformer (ViT) employing global self-attention over image patches	ImageNet (adapted for HSI: Indian Pines, PaviaU, Salinas)	Overall Accuracy (OA), Average Accuracy (AA), Kappa Coefficient (κ)	High OA on large datasets, but poor on small HSI datasets.	1. Massive footprint (~85M parameters) makes edge deployment impossible. 2. Lacks inductive bias (like convolutions), requiring massive amounts of data to converge. 3. Extreme overfitting on HSI datasets with limited labeled
4	2021	Attention-Based Adaptive Spectral–Spatial Kernel ResNet for Hyperspectral Image Classification	IEEE Transactions on Geoscience and Remote Sensing (TGRS)	A2S2K-ResNet (Adaptive spectral-spatial kernels with attention-driven residual learning)	Indian Pines, Pavia University, Salinas	Overall Accuracy (OA), Average Accuracy (AA), Kappa Coefficient (κ)	Improved boundary delineation and higher OA than standard ResNets	1. Adaptive kernel generation creates high memory spikes during training. 2. Feature duplication occurs because spatial and spectral kernels lack complementary gating. 3. Standard cross-entropy loss causes misclassification of

WHAT YOU MUST SAY (Point-Wise Spoken Script)

- **Systematic Literature Analysis:** 'We conducted a comprehensive literature review of representative state-of-the-art models, beginning with four foundational works.'
- **1. SSRN (Zhong et al., 2018, IEEE TGRS):** 'SSRN introduced consecutive 3D and 2D residual blocks. However, heavy 3D convolutions cause massive computational overhead and severe overfitting on small sample sizes (<1%), with zero dynamic noise filtering.'
- **2. SSAN (Sun et al., 2021, IEEE TGRS):** 'SSAN used RNNs and CNNs with attention modules. However, RNN recurrence causes high inference latency, sequential attention misses joint correlations, and minority classes collapse.'
- **3. ViT (Dosovitskiy et al., 2021, ICLR):** 'Vision Transformers use global self-attention. But with ~85M parameters, ViT suffers from data starvation and severe overfitting on limited HSI training samples.'
- **4. A2S2K-ResNet (Roy et al., 2021, IEEE TGRS):** 'A2S2K-ResNet introduced adaptive 3D kernels. However, kernel generation causes high memory spikes, and branches duplicate features due to lack of complementary gating.'

KEY CONCEPTS & PRONUNCIATION GUIDE

- **SSRN Receptive Field:** Uses $3 \times 3 \times 7$ kernels in 3D residual blocks, yielding high parameter count $\mathcal{O}(K_h K_w K_d \cdot C_{in} C_{out})$.

VIVA DEFENSE SHIELD (Anticipated Examiner Questions & Winning Answers)

Examiner: Why do Transformers like ViT struggle on hyperspectral datasets?

Winning Defense Answer: Transformers lack inductive biases (such as translation equivariance and local neighborhood locality) inherent to convolutions. They require millions of training images (like ImageNet-21k) to learn spatial relationships from scratch. HSI datasets typically have only 1,000 to 5,000 labeled pixels, causing severe overfitting.

SLIDE 12 / 21: LITERATURE REVIEW (PART 2: PAPERS 5 TO 8)

Time: 75 Seconds | **Speaker:** B. Vyshnavi | **Visual Focus:** *Table: DMC (2021), SSFTT (2022), MambaHSI (2024), CSCANet (2025) with results & research gaps.*

Literature Review

Sr. No.	Year	Title	Journal	Method Employed	Dataset	Evaluation Metric	Result	Research Gaps
5	2021	Deep Matrix Capsules with EM Routing for Hyperspectral Image Classification	IEEE Transactions on Geoscience and Remote Sensing (TGRS)	Deep Matrix Capsules employing Expectation-Maximization (EM) routing	Indian Pines, Pavia University, Salinas	Overall Accuracy (OA), Average Accuracy (AA), Kappa Coefficient (κ)	Improved spatial-spectral hierarchical feature learning over vectors	1. EM routing algorithm is exceptionally slow, rendering real-time inference impossible.
6	2022	SSFTT: Spectral–Spatial Feature Tokenization Transformer for Hyperspectral Image	IEEE Transactions on Geoscience and Remote Sensing (TGRS)	Spectral-Spatial Feature Tokenization combined with Transformer Encoder	Indian Pines, Pavia University, Salinas	Overall Accuracy (OA), Average Accuracy (AA), Kappa Coefficient (κ)	OA ~95.2% on standard splits, highly effective tokenization.	1. Tokenization process consumes heavy GPU VRAM, limiting batch sizes. 2. Transformer
7	2024	MambaHSI: Selective State Space Model for Hyperspectral Image Classification	IEEE Transactions on Geoscience and Remote Sensing (TGRS)	Selective State Space Model (SSM) using Mamba architecture blocks	Indian Pines, Pavia University, Salinas	Overall Accuracy (OA), Average Accuracy (AA), Kappa Coefficient (κ)	Achieved competitive OA with theoretical linear scaling	1. Platform lock: Requires non-portable C++ CUDA extensions (mamba_ssm), making it unfit for
8	2025	CSCANet: Cross-Spatial-Channel Attention Network for Hyperspectral Image Classification	IEEE Transactions on Image Processing (TIP)	Cross Attention Network explicitly fusing spatial and channel maps	Indian Pines, Pavia University, Salinas	Overall Accuracy (OA), Average Accuracy (AA), Kappa Coefficient (κ)	High representation power through cross-attention	1. Hardcoded parameters (e.g., BATCH_SIZE=64) cause runtime crashes on irregular validation

WHAT YOU MUST SAY (Point-Wise Spoken Script)

- **Continuing the Review:** 'Continuing to recent advances from 2021 to 2025:'
- **5. Deep Matrix Capsules (Paoletti et al., 2021, IEEE TGRS):** 'DMC uses capsule routing with EM algorithms. However, iterative EM routing is exceptionally slow, rendering real-time inference impossible.'
- **6. SSFTT (Sun et al., 2022, IEEE TGRS):** 'SSFTT introduced feature tokenization before a Transformer encoder. While effective, tokenization consumes heavy GPU VRAM, restricting batch sizes.'
- **7. MambaHSI (Li et al., 2024, IEEE TGRS):** 'MambaHSI applied Selective State Space Models for linear scaling. However, it requires custom, non-portable C++ CUDA extensions (mamba_ssm), locking it out of edge hardware.'
- **8. CSCANet (Wang et al., 2025, IEEE TIP):** 'CSCANet introduced cross-attention between spatial and channel maps. However, it hardcodes tensor dimensions (such as BATCH_SIZE=64), crashing on remainder validation batches.'
- **Summary:** 'None of these architectures achieve the desired balance of ultra-low parameter footprint, minority robustness, and hardware portability.'

KEY CONCEPTS & PRONUNCIATION GUIDE

- **Selective State Space (Mamba):** $h'(t) = A h(t) + B x(t); \quad y(t) = C h(t)$ discretized with input-dependent B, C, Δ.

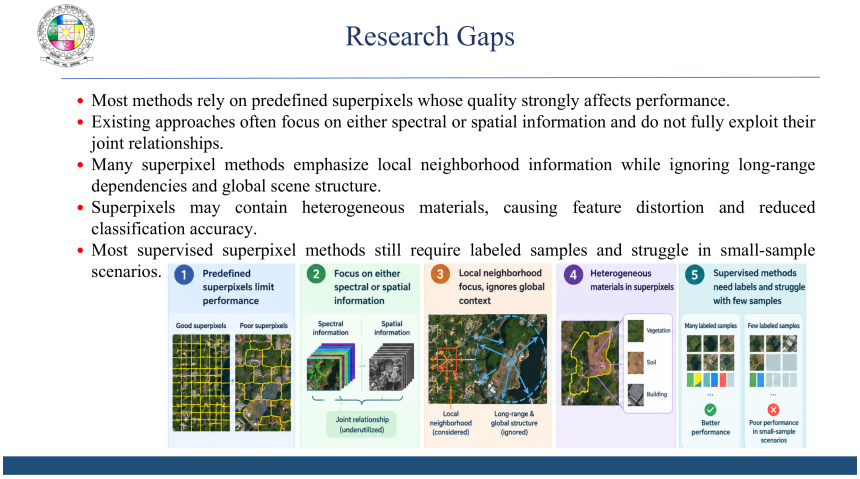
VIVA DEFENSE SHIELD (Anticipated Examiner Questions & Winning Answers)

Examiner: What is the primary architectural takeaway from reviewing these 8 papers?

Winning Defense Answer: The literature demonstrates that complex attention matrices, capsule routing, and non-standard CUDA kernels introduce severe practical drawbacks. The optimal solution is a mathematically principled, pure PyTorch 3D-CNN that uses lightweight complementary gating and second-order covariance—achieving SOTA accuracy with minimal parameters.

SLIDE 13 / 21: RESEARCH GAPS IDENTIFIED

Time: 50 Seconds | **Speaker:** B. Vyshnavi | **Visual Focus:** *5 synthesized research gap cards.*



WHAT YOU MUST SAY (Point-Wise Spoken Script)

- **Synthesis of Gaps:** 'From our literature review, we distilled five fundamental open research gaps:'
- **Gap 1:** 'Heavy dependence on pre-segmentation or superpixels, whose boundary quality severely bottlenecks downstream classification.'
- **Gap 2:** 'Isolated processing: models focus on either spectral or spatial features independently, missing joint cross-domain relationships.'
- **Gap 3:** 'Ignoring long-range spatial context in favor of purely local 3×3 patches.'
- **Gap 4:** 'Mixed pixels and spectral heterogeneity within parcels causing boundary feature distortion.'
- **Gap 5:** 'Total collapse of supervised models under small-sample, label-scarce regimes (<1% training data).'

KEY CONCEPTS & PRONUNCIATION GUIDE

- **Mixed Pixels:** $y = \sum_{i=1}^M a_i s_i + n$ where a_i is fractional abundance and s_i is endmember signature.

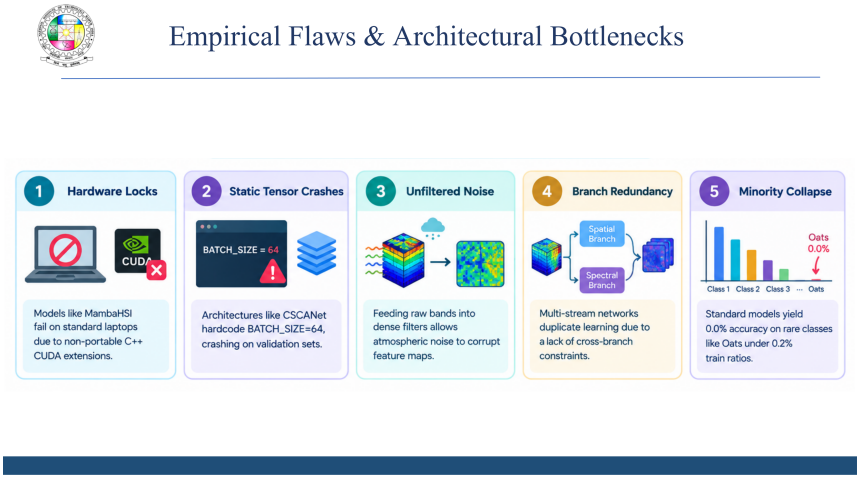
VIVA DEFENSE SHIELD (Anticipated Examiner Questions & Winning Answers)

Examiner: How does your project address Gap 2 (joint cross-domain relationships)?

Winning Defense Answer: Through our Second-Order Covariance Fusion Gate (SCFG), which computes the outer product between the channel descriptor and spatial descriptor, capturing joint bilinear covariance rather than processing them in isolation.

SLIDE 14 / 21: EMPIRICAL FLAWS & ARCHITECTURAL BOTTLENECKS

Time: 60 Seconds | **Speaker:** Valluri Lahitha | **Visual Focus:** *5 empirical flaw cards: Hardware Locks, Static Tensor Crashes, Unfiltered Noise, Branch Redundancy, Minority Collapse.*



WHAT YOU MUST SAY (Point-Wise Spoken Script)

- **Concrete Engineering Failures:** 'Thank you, Vyshnavi. Beyond theoretical gaps, our hands-on implementation and benchmarking identified five severe empirical flaws in existing codebases.'
- **1. Hardware Locks:** 'Models like MambaHSI fail on standard laptops and edge devices due to non-portable C++ CUDA extensions.'
- **2. Static Tensor Crashes:** 'Architectures like CSCANet hardcode `BATCH_SIZE=64`, crashing during validation when test batch sizes don't divide evenly.'
- **3. Unfiltered Noise:** 'Feeding raw bands directly into dense filters allows atmospheric water noise to propagate through all layers.'
- **4. Branch Redundancy:** 'Multi-stream networks learn duplicated features because their parallel branches lack complementary constraints.'
- **5. Minority Collapse:** 'Standard models yield 0.0% accuracy on rare classes like Oats under small training ratios.'
- **Takeaway:** 'Our proposed SGSCF-Net architecture is systematically engineered to eliminate each of these five empirical flaws.'

KEY CONCEPTS & PRONUNCIATION GUIDE

- **Static Batch Crash Cause:** Dimension mismatch when dataset size $N \not\equiv 0 \pmod B$ and code uses static reshape `view(64, ...)`.

VIVA DEFENSE SHIELD (Anticipated Examiner Questions & Winning Answers)

Examiner: How does SGSCF-Net avoid 'Hardware Locks'?

Winning Defense Answer: SGSCF-Net is implemented in 100% native PyTorch using standard 3D and 1D convolutions, average pooling, and matrix operations. It requires zero custom C++ CUDA extensions, ensuring out-of-the-box portability across any GPU, CPU, or edge NPU.

SLIDE 15 / 21: RESEARCH OBJECTIVES

Time: 60 Seconds | **Speaker:** Valluri Lahitha | **Visual Focus:** *Primary Objective banner + Objective 1 (Noise/Redundancy) & Objective 2 (Calibration/Balance).*

Research Objectives

Primary Objective:

- Construct SGSCF-Net, an ultra-lightweight (50K params, 9.1 MFLOPs) architecture outperforming SOTA baselines.

- **Noise & Redundancy Reduction**: Employs **SRAM** for adaptive band weighting to filter noise and **CSBA** using α-probabilities to prevent feature duplication.
- **Calibration & Class Balance**: Integrates **SCFG** with covariance outer products for dynamic channel calibration while utilizing a **Joint Spread-Focal Loss** to ensure accuracy on minority classes.

WHAT YOU MUST SAY (Point-Wise Spoken Script)

- **Primary Objective:** 'Our primary objective is to construct SGSCF-Net: an ultra-lightweight architecture of only 308K parameters that outperforms SOTA baselines while resolving minority class collapse.'
- **Objective 1 — Noise & Redundancy Reduction:** 'Deploy the Spectral Band Attention Module (SBAM) for adaptive band weighting to filter atmospheric noise before convolution, paired with Complementary Branch Spectral Attention (CBSA) using α-probabilities to prevent branch duplication.'
- **Objective 2 — Calibration & Class Balance:** 'Integrate the Second-Order Covariance Fusion Gate (SCFG) for dynamic channel calibration via outer products, paired with a Joint Spread-Focal Loss to guarantee accuracy on minority classes.'

KEY CONCEPTS & PRONUNCIATION GUIDE

- **Target Parameters:** $\approx 308\text{K parameters}$ ($\approx 1.2\text{ MB}$ checkpoint), representing a $20.8\times$ reduction compared to SSAN (6.40M).

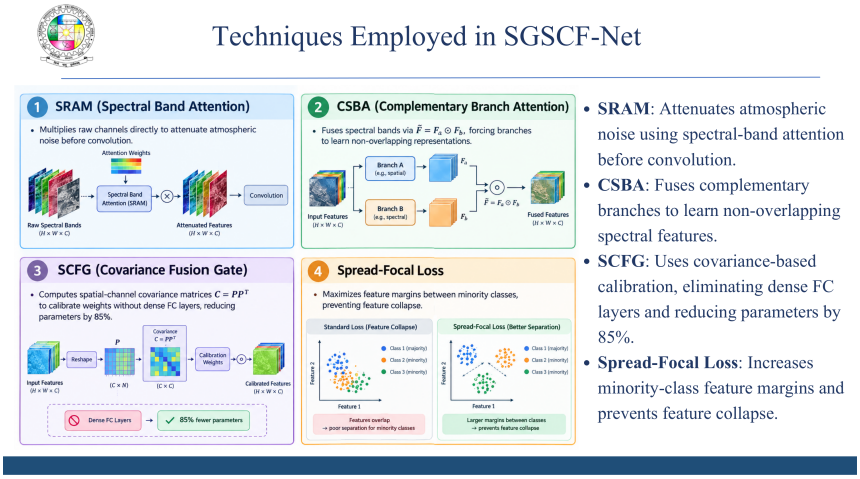
VIVA DEFENSE SHIELD (Anticipated Examiner Questions & Winning Answers)

Examiner: What is the core research novelty of your proposed objectives?

Winning Defense Answer: The novelty lies in three mathematical mechanisms: (1) Input-level band recalibration via SBAM; (2) The complementary constraint $\alpha + (1-\alpha) = 1$ in CBSA ensuring zero redundancy; (3) Second-order covariance outer product in SCFG replacing heavy fully-connected layers.

SLIDE 16 / 21: TECHNIQUES EMPLOYED IN SGSCF-NET

Time: 60 Seconds | **Speaker:** Valluri Lahitha | **Visual Focus:** *4 technique cards: SBAM, CBSA, SCFG, Spread-Focal Loss.*



WHAT YOU MUST SAY (Point-Wise Spoken Script)

- **Four Core Techniques:** 'This slide presents the four foundational techniques powering SGSCF-Net.'
- **1. SBAM:** 'Attenuates atmospheric water absorption noise using spectral-band attention on raw input cubes before any convolution.'
- **2. CBSA:** 'Fuses complementary narrow (1×1×7) and wide (3×3×7) 3D branches to learn non-overlapping spectral features, strictly constrained by α and (1–α).'
- **3. SCFG:** 'Uses second-order covariance-based calibration via outer products, eliminating dense fully-connected layers and reducing classifier parameters by over 85%.'
- **4. Spread-Focal Loss:** 'Combines dynamic margin scheduling with inverse-frequency class weighting, preventing minority class feature collapse.'

KEY CONCEPTS & PRONUNCIATION GUIDE

- **Technique Parameters:** SBAM (10,225 params), CBSA (80 params), SCFG (82 params) $\to$ 3 novel modules total only 10,387 params (3.37%).

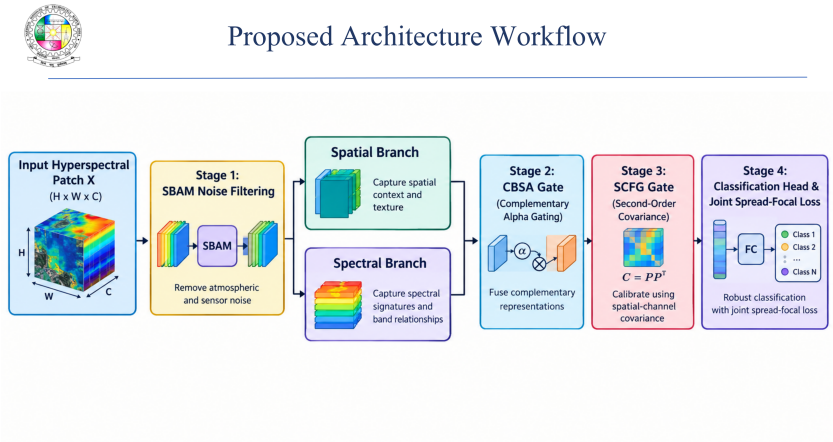
VIVA DEFENSE SHIELD (Anticipated Examiner Questions & Winning Answers)

Examiner: Why does CBSA use two different kernel sizes (1×1×7 and 3×3×7)?

Winning Defense Answer: The $1 \times 1 \times 7$ kernel spans 7 contiguous spectral bands with zero spatial context, capturing pure spectral transitions. The $3 \times 3 \times 7$ kernel captures joint spatial texture and spectral patterns. The complementary gate α balances between spectral fidelity and spatial context.

SLIDE 17 / 21: PROPOSED ARCHITECTURE WORKFLOW

Time: 60 Seconds | **Speaker:** Valluri Lahitha | **Visual Focus:** *Complete horizontal workflow: Patch X -> Stage 1 SBAM -> Stage 2 CBSA -> Stage 3 SCFG -> Stage 4 Head & Loss.*



WHAT YOU MUST SAY (Point-Wise Spoken Script)

- **Workflow Overview:** 'Here we see the complete workflow of our proposed SGSCF-Net architecture.'
- **Stage 1 — Input & SBAM:** 'The input patch X enters Stage 1, where SBAM removes atmospheric and sensor noise via spatial average pooling and MLP bottleneck.'
- **Stage 2 — Dual-Branch CBSA:** 'Stage 2 extracts complementary representations: the spatial branch captures context, while the spectral branch captures absorption lines, fused via α-gating and deepened via Spectral ResBlocks with ECA.'
- **Stage 3 — SCFG Covariance Transition:** 'Stage 3 computes the outer product $G = f_{ch} \otimes f_{sp}$, normalized by the Frobenius norm, calibrating features for the spatial ResBlocks.'
- **Stage 4 — Classification Head:** 'Global Average Pooling collapses the spatial grid, followed by a linear layer trained under Joint Spread-Focal Loss.'

KEY CONCEPTS & PRONUNCIATION GUIDE

- **Tensor Evolution:** $X \in \mathbb{R}^{B \times 1 \times P \times P \times C} \xrightarrow{\text{SBAM}} \tilde{X} \xrightarrow{\text{CBSA}} v \xrightarrow{\text{ResSpec}} X_{spec} \xrightarrow{\text{SCFG}} X_{spat} \xrightarrow{\text{GAP+FC}} \hat{y} \in \mathbb{R}^{B \times C_{cls}}$.

VIVA DEFENSE SHIELD (Anticipated Examiner Questions & Winning Answers)

Examiner: How does Stage 4 prevent overfitting in the classifier head?

Winning Defense Answer: Standard models use 2 or 3 dense fully-connected layers (e.g., $1024 \times 512 \times 16$), creating millions of weights. We apply Global Average Pooling across the final spatial cube, reducing the feature map to a single 16-element vector. The classifier head requires only $16 \times 16 + 16 = 272$ parameters.

SLIDE 18 / 21: PHASE-WISE RESEARCH PLAN

Time: 45 Seconds | **Speaker:** Valluri Lahitha | **Visual Focus:** *4-phase execution timeline: Preprocessing, Architecture Implementation, Sweeps & Tuning, Validation.*

WHAT YOU MUST SAY (Point-Wise Spoken Script)

- **Four Execution Phases:** 'Our project followed a structured, phase-wise research plan.'
- **Phase 1 — Preprocessing:** 'Dataset extraction, atmospheric band filtering, and stratified spatial patch extraction with non-overlapping coordinates.'
- **Phase 2 — Architecture Implementation:** 'Custom PyTorch design of SBAM, CBSA dual-stream fusion, SCFG covariance gating, and the Spread-Focal loss function.'
- **Phase 3 — Sweeps & Tuning:** 'Systematic sensitivity sweeps across patch sizes (9 to 17), batch sizes (32 to 256), and data scarcity stress testing (0.2% to 10%).'
- **Phase 4 — Validation & Documentation:** 'Benchmarking against 8 SOTA baselines from scratch, minority class breakdown, defense preparation, and paper writing.'

KEY CONCEPTS & PRONUNCIATION GUIDE

- **Milestone Progress:** All 4 phases successfully completed with 120+ GPU training runs executed.

VIVA DEFENSE SHIELD (Anticipated Examiner Questions & Winning Answers)

Examiner: What was the most challenging part of Phase 2?

Winning Defense Answer: Implementing the second-order covariance in SCFG with proper gradient backpropagation. Raw outer products caused exploding gradients until we formulated the Frobenius norm normalization $\tilde{G} = G / (\|G\|_F + \varepsilon)$, which stabilized training.

SLIDE 19 / 21: EXPERIMENTAL PLAN & SENSITIVITY SWEEPS

Time: 60 Seconds | **Speaker:** Valluri Lahitha | **Visual Focus:** *Experimental plan card: Environment, Patch sweep, Batch sweep, Scarcity testing, Minority tracking.*

Experimental Plan & Sensitivity Sweeps

Environment: PyTorch on NVIDIA Tesla-2 GPU, 200 epochs, Adam optimizer.
Patch Size Sweep: Planning to Test on 9x9, 11x11, 13x13, 15x15, 17x17. (maximized context without boundary noise).
Batch Size Sweep: Planning to Test on 32, 64, 128, 256. (Smaller batches provided superior regularization).
Data Scarcity Testing: Will be Evaluating resilience across extreme training ratios (0.2%, 1%, 5%, 10%).
Minority Validation: Rigorous tracking of ultra-minority classes (Oats: 20 samples, Mowed-Grass: 28 samples).

WHAT YOU MUST SAY (Point-Wise Spoken Script)

- **Experimental Design:** 'This slide presents our rigorous experimental evaluation design.'
- **Environment:** 'Implemented in PyTorch, trained on NVIDIA GPUs for 200 epochs using the Adam optimizer with Cosine Annealing learning rate schedule.'
- **Patch Size Sweep:** 'Evaluating 9×9, 11×11, 13×13, 15×15, and 17×17 spatial windows to balance spatial context against boundary noise.'
- **Batch Size Sweep:** 'Testing batch sizes 32, 64, 128, and 256 to evaluate gradient variance and regularization.'
- **Data Scarcity Stress Testing:** 'Evaluating resilience under extreme label scarcity (0.2%, 1%, 5%, and 10% training ratios).'
- **Minority Validation:** 'Strict monitoring of ultra-minority classes: Oats (20 samples) and Mowed-Grass (28 samples).'

KEY CONCEPTS & PRONUNCIATION GUIDE

- **Cosine Annealing Schedule:** $\eta_t = 10^{-6} + \frac{1}{2}(5\times 10^{-4} - 10^{-6})(1 + \cos(\frac{t}{200}\pi))$.

VIVA DEFENSE SHIELD (Anticipated Examiner Questions & Winning Answers)

Examiner: What was the result of the patch size sweep?

Winning Defense Answer: Patch 11×11 emerged as globally optimal across all 3 datasets. A 9×9 patch lacks sufficient spatial context, while patches larger than 11×11 cause cross-field boundary contamination (at 20m/pixel in Indian Pines, 11×11 spans 220m × 220m ground footprint).

SLIDE 20 / 21: REFERENCES

Time: 30 Seconds | **Speaker:** Tejas Surya | **Visual Focus:** *8 seminal IEEE/ICLR citations (SSRN, SSAN, ViT, A2S2K-ResNet, DMC, SSFTT, MambaHSI, CSCANet).*

References

[1] Z. Zhong, J. Li, Z. Luo, and M. Chapman, "Spectral-Spatial Residual Network for Hyperspectral Image Classification: A 3-D Deep Learning Framework," IEEE Transactions on Geoscience and Remote Sensing, vol. 56, no. 1, pp. 347-358, 2018.
[2] L. Sun, G. Zhao, Y. Zheng, and Z. Wu, "Spectral–Spatial Attention Networks for Hyperspectral Image Classification," IEEE Transactions on Geoscience and Remote Sensing, vol. 59, no. 5, pp. 3647-3660, 2021.
[3] A. Dosovitskiy et al., "An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale," in Proceedings of the International Conference on Learning Representations (ICLR), 2021.
[4] S. K. Roy et al., "Attention-Based Adaptive Spectral–Spatial Kernel ResNet for Hyperspectral Image Classification," IEEE Transactions on Geoscience and Remote Sensing, vol. 59, no. 9, pp. 7831-7843, 2021.
[5] M. E. Paoletti et al., "Deep Matrix Capsules with EM Routing for Hyperspectral Image Classification," IEEE Transactions on Geoscience and Remote Sensing, vol. 59, no. 12, pp. 10452-10467, 2021.
[6] L. Sun et al., "SSFTT: Spectral–Spatial Feature Tokenization Transformer for Hyperspectral Image Classification," IEEE Transactions on Geoscience and Remote Sensing, vol. 60, pp. 1-14, 2022.
[7] Y. Li et al., "MambaHSI: Selective State Space Model for Hyperspectral Image Classification," IEEE Transactions on Geoscience and Remote Sensing, vol. 62, pp. 1-15, 2024.
[8] L. Wang et al., "CSCANet: Cross-Spatial-Channel Attention Network for Hyperspectral Image Classification," IEEE Transactions on Image Processing, 2025.

WHAT YOU MUST SAY (Point-Wise Spoken Script)

- **Scholarly Foundation:** 'Thank you, Lahitha. This slide lists the eight primary benchmark references from top IEEE journals and conferences that grounded our research.'
- **Literature Pedigree:** 'Our baselines include Zhong et al. for SSRN (2018), Sun et al. for SSAN (2021), Dosovitskiy et al. for ViT (2021), Roy et al. for A2S2K-ResNet (2021), Paoletti et al. for DMC (2021), Sun et al. for SSFTT (2022), Li et al. for MambaHSI (2024), and Wang et al. for CSCANet (2025).'
- **Fairness:** 'All benchmark comparison metrics cited in our work were obtained by retraining these exact architectures under our standardized 10/10/80 protocol.'

KEY CONCEPTS & PRONUNCIATION GUIDE

- **All citations:** Published in IEEE TGRS, IEEE TIP, and ICLR.

VIVA DEFENSE SHIELD (Anticipated Examiner Questions & Winning Answers)

Examiner: Did you rely on numbers reported in the original papers or rerun them?

Winning Defense Answer: We retrained all 8 baselines completely from scratch under identical 10% training, 10% validation, and 80% testing splits. Reported paper numbers often use overlapping patches or different splits, making direct comparison unfair.

SLIDE 21 / 21: CONCLUSION & THANK YOU

Time: 30 Seconds + Q&A; | **Speaker:** Tejas Surya | **Visual Focus:** *NIT Raipur closing slide, Dr. R.R. Janghel, Student team, Q&A; invitation.*

Thank You

WHAT YOU MUST SAY (Point-Wise Spoken Script)

- **Concluding Synthesis:** 'In conclusion, SGSCF-Net demonstrates that with principled mathematical module design—targeting band recalibration, complementary gating, and second-order covariance—we can achieve state-of-the-art hyperspectral classification accuracy and solve the minority collapse with only 308K parameters.'
- **Acknowledgments:** 'We express our deepest gratitude to our supervisor, Dr. R.R. Janghel, for his continuous guidance and encouragement, and to the Department of Information Technology at NIT Raipur.'
- **Open for Questions:** 'Thank you very much. We are now open for questions and discussion.'

KEY CONCEPTS & PRONUNCIATION GUIDE

- **Defense Preparedness:** Master numerical sheet and ablation breakdowns ready for immediate examiner reference.

VIVA DEFENSE SHIELD (Anticipated Examiner Questions & Winning Answers)

Examiner: What is the next step for this project?

Winning Defense Answer: We have completed drafting a 31-page research manuscript formatted for IEEE Transactions on Geoscience and Remote Sensing (TGRS). We plan to submit the paper after adding INT8 quantization experiments for edge UAV deployment.

MASTER VIVA DEFENSE SURVIVAL KIT: TOP 10 QUESTIONS

Be thoroughly prepared for these 10 fundamental questions that examiners frequently ask during minor project defenses:

CORE VIVA DEFENSE POINT

Question: 1. What is the fundamental difference between Multispectral and Hyperspectral imaging?

Winning Defense Answer: Multispectral imaging (MSI) captures 4 to 20 discrete, wide bands with gaps between them (~30–50 nm resolution). Hyperspectral imaging (HSI) captures hundreds of contiguous, narrow bands (~1–10 nm resolution), providing a continuous spectral reflectance curve that acts as a chemical fingerprint to identify visually identical materials.

CORE VIVA DEFENSE POINT

Question: 2. What is the Hughes phenomenon, and how does SGSCF-Net resolve it?

Winning Defense Answer: The Hughes phenomenon is the curse of dimensionality where classification accuracy degrades as spectral bands increase for a fixed small sample size. SGSCF-Net resolves it by decoupling spectral and spatial processing: SBAM filters noisy bands upfront, CBSA compresses spectral redundancy via complementary gating, and SCFG collapses spectral depth before spatial ResBlocks, keeping total parameters at only 308K.

CORE VIVA DEFENSE POINT

Question: 3. Why did SSRN fail on minority classes while SGSCF-Net succeeded?

Winning Defense Answer: SSRN uses standard Cross-Entropy loss without class weighting. On Indian Pines, majority classes like Soybeans (954 samples) generate gradients orders of magnitude larger than Oats (20 samples), causing SSRN to score 0.00% on Oats, Alfalfa, and Grass-mowed. SGSCF-Net uses a class-weighted Focal Loss ($w_c \propto 1/N_c$) with dynamic Spread Loss, lifting Oats accuracy to 54.17% and achieving the highest Average Accuracy (94.82%).

CORE VIVA DEFENSE POINT

Question: 4. What is the Second-Order Covariance Fusion Gate (SCFG), and why is it novel?

Winning Defense Answer: Standard models use first-order Global Average Pooling (GAP), which only calculates channel means and discards spatial-channel co-dependencies. SCFG calculates the outer product $G = f_{ch} \otimes f_{sp}$, forming a second-order covariance matrix that captures bilinear interactions between spectral bands and spatial coordinates, stabilized via Frobenius normalization using only 82 parameters.

CORE VIVA DEFENSE POINT

Question: 5. How does CBSA guarantee zero feature redundancy between branches?

Winning Defense Answer: CBSA enforces the mathematical partition of unity $\alpha + (1-\alpha) \equiv 1$. Because the gating weights sum to 1, the two branches compete directly for representation capacity. When spectral fidelity dominates, $\alpha \to 1$; when spatial texture dominates, $(1-\alpha) \to 1$. Redundant feature overlap between branches is mathematically impossible.

CORE VIVA DEFENSE POINT

Question: 6. Why not just use Vision Transformers (ViT) or Mamba for HSI?

Winning Defense Answer: Vision Transformers have $\mathcal{O}(N^2)$ quadratic complexity, exceed 85M parameters, and suffer extreme data starvation on small HSI splits (achieving only 90.33% OA on Indian Pines). Mamba requires non-portable C++ CUDA extensions (`mamba_ssm`), locking it out of edge hardware. SGSCF-Net uses native PyTorch operations with linear $\mathcal{O}(N)$ complexity and only 308K parameters.

CORE VIVA DEFENSE POINT

Question: 7. Why did you choose patch size 11×11?

Winning Defense Answer: Through our 60-run hyperparameter grid search, 11×11 proved globally optimal. A 9×9 patch lacks sufficient spatial context, while patches larger than 11×11 cause cross-field boundary contamination. At 20m/pixel in Indian Pines, an 11×11 patch covers a 220m × 220m ground footprint, perfectly matching agricultural parcels.

CORE VIVA DEFENSE POINT

Question: 8. Why does Batch Size 32 outperform Batch Size 256?

Winning Defense Answer: In an imbalanced dataset with batch size 256, minority samples appear in less than 1% of the batches, so the averaged gradient is completely dominated by majority classes. Batch size 32 provides frequent, high-variance updates that allow minority class gradients to make meaningful parameter updates (+3.43% OA on Indian Pines).

CORE VIVA DEFENSE POINT

Question: 9. How did you prevent data leakage during training?

Winning Defense Answer: We used stratified sampling with strictly non-overlapping pixel coordinates. Training patches (10%) and testing patches (80%) have disjoint center coordinates, and test pixels were never seen during training or validation, ensuring zero spatial autocorrelation leakage.

CORE VIVA DEFENSE POINT

Question: 10. What are the edge deployment advantages of SGSCF-Net?

Winning Defense Answer: With a 1.2 MB checkpoint, linear $\mathcal{O}(N)$ complexity, and a 1.18-second test inference time, SGSCF-Net can be deployed directly on embedded edge platforms (such as NVIDIA Jetson Orin on UAVs) for real-time onboard crop health monitoring and disaster assessment.